BChemXtract 1.4.1 vs ChemScanner, RDKit and Indigo

Structure-extraction performance over the BChemXtract ChemDraw test corpus, with Open Babel as the baseline from the previous report

Generated 2026-10-15 · 225 binary `.cdx` + 21 `.cdxml` files under `testset/corpus` · BChemXtract 1.4.1 (tag `v1.4.1`) and current `main` (`f1db7bf`), JDK 25 vs ChemScanner 0.1.3, RDKit 2026.09.1, Indigo 1.46.0, Open Babel 3.1.1 · identical corpus, all tools deduplicating per file on InChIKey, all runs on one machine on the same day

At a glance

CHEMICAL QUALITY
98–99% vs 73–77%
Real InChIs: BChemXtract (R-groups off/on) vs ChemScanner, Indigo, RDKit (`.cdx`)

UNEXPANDED ALIASES
18–25 vs 298–378
Leftover `*` atoms in the unique set

ROBUSTNESS
1 vs 0–33
Failed files of 225; the 1 is a non-CDX file every reader rejects

SPEED
36 vs 25–64 ms
Per file; Indigo is fastest, slow outliers excluded for others

Comparison on the 225 `.cdx` files

METRIC	BCHEMXTRACT 1.4.1	BCHEMXTRACT 1.4.1 + R-GROUPS	BCHEMXTRACT MAIN + R-GROUPS F1DB7BF	CHEMSCANNER 0.1.3	RDKIT 2026.09.1	INDIGO 1.46.0	OPEN BABEL 3.1.1
COVERAGE							
Files yielding structures	203	210	210	192	218	219	224
Empty (0 found)	21	14	14	0	3	6	0
Failures	1	1	1	33	4 (timeouts)	**0**	1
YIELD & CHEMICAL QUALITY							
Unique substances	1360	1771	1797	1310	1515	1490	1317
Real InChIs generated	1336 (98.2%)	1752 (98.9%)	**1777 (98.9%)**	956 (73.0%)	1164 (76.8%)	1106 (74.2%)	187 (14.2%)
SMILES fallback (no InChI)	24	**19**	20	354	351	384	1130
Structures with unexpanded alias (`*`)	23	**18**	19	298	334	378	617
Fully expanded structures	1337	1753	**1778**	1012	1181	1112	700
Raw structures before dedup	1979 fragments	1979 fragments	1998 fragments	1835	1925	1911	1864
SPEED							
Total time	8.1 s	8.6 s	10.8 s	34.0 s	555.4 s	**5.6 s**	72.9 s
Average per file	36 ms	38 ms	48 ms	151 ms	2468 ms	**25 ms**	324 ms
Average per file, slow outliers excluded	36 ms (none)	38 ms (none)	34 ms (1 file)	64 ms (2 files)	51 ms (7 files)	**25 ms (none)**	75 ms (1 file)
Average per file, warm	18 ms	23 ms	35 ms	100 ms	247 ms	**5.8 ms**	n/a

"+ R-groups" runs the same extraction with `resolveRGroups = true`, which substitutes R-groups defined on the page and enumerates Markush structures. Its extra substances are mostly enumerated Markush members, so its "unique substances" count is not like-for-like with the other readers; the real-InChI share and the `*` count compare fine. The plain columns use `false`, as in the 2026-09-09 report. "main + R-groups" is the head of `main` at `f1db7bf` (after 1.5.0, untagged), run with `true`. The warm row loops over the corpus three times in one process per tool and reports the third pass. It uses the 221 files left after dropping the 4 on which RDKit hangs, and RDKit's figure still includes two files that take about 53 s and 10 s. ChemScanner's figure excludes InChI generation, which its Ruby binding cannot do. Open Babel was run as a command-line tool and has no in-process figure. "Slow outliers" are files over 1 s. BChemXtract's slowest file takes 0.64 s and Indigo's 0.10 s, so neither has any. Failures include timeouts (120 s cap per file).

> **RDKit's total is four hung files.** RDKit does not finish `bjnano_5_145/m5445236-i4.cdx`, `-i13.cdx`, `m16284363-12.cdx` and `oome.cdx` within 120 s; with three more slow files from the same `volumeTest/bug` set they take 544 of its 555 s. The hang is in reading, not in InChI generation. On three of the files, RDKit's CDX reader with sanitisation on (its default) does not finish, although reading without sanitisation and sanitising afterwards takes under a second. On `m5445236-i13` even the unsanitised read stalls, and one of its structures also hangs RDKit's aromaticity perception when parsed from SMILES. On the other 218 files RDKit averages 51 ms.

Comparison on the 21 `.cdxml` files

METRIC	BCHEMXTRACT 1.4.1	BCHEMXTRACT 1.4.1 + R-GROUPS	BCHEMXTRACT MAIN + R-GROUPS F1DB7BF	CHEMSCANNER 0.1.3	RDKIT 2026.09.1	INDIGO 1.46.0
Files yielding structures	17	20	20	9	19	14
Empty (0 found) / failures	4 / 0	1 / 0	1 / 0	0 / 12	2 / 0	0 / 7
Unique substances	55	86	74	249	412	55
Real InChIs	51	84	72	34	392	35
Unexpanded alias (`*`)	4	2	2	215	20	19
EXCLUDING `volumeTest/bug/Figure1C_complete_curated.cdxml`						
Unique substances	55	86	74	43	73	55
Real InChIs	51 (92.7%)	**84 (97.7%)**	72 (97.3%)	34 (79.1%)	53 (72.6%)	35 (63.6%)
Unexpanded alias (`*`)	4	**2**	**2**	9	20	19
Total time	2.9 s	1.7 s	1.6 s	5.1 s	1.0 s	**0.3 s**

Open Babel 3.1.1 has no CDXML reader and is not in this table. Figure1C is listed separately because one file dominates the totals: RDKit reads about 340 sugar building blocks from it (339 InChIs), and BChemXtract returns none.

Key findings

- **Chemistry is still the decisive difference, but the gap is far smaller than against Open Babel.** BChemXtract produces a real InChI for 98.2% of its unique `.cdx` substances. RDKit (76.8%), Indigo (74.2%) and ChemScanner (73.0%) are much closer to that than Open Babel (14.2%), but each still leaves 298–378 structures carrying `*` atoms. Sampled, these are labels the readers keep as dummy atoms: R-group placeholders (`[*:1]`), nicknames on metals and phosphines (`[Cu]*`), and label-only fragments (`*.CC(=O)[O][K]`). BChemXtract resolves most of them into atoms.
- **The higher raw counts of RDKit and Indigo are not extra chemistry.** They report 1515 and 1490 unique substances against BChemXtract's 1360, but 334 and 378 of these are wildcard structures. In fully expanded structures BChemXtract leads at 1337, ahead of RDKit's 1181 and Indigo's 1112.
- **Resolving R-groups closes most of BChemXtract's empty files.** With `resolveRGroups = true`, 7 of the 21 empty `.cdx` files yield structures (the Markush fixtures plus `m5445236-i2` and `m30177584-2`), and real InChIs rise to 1752 of 1771 (98.9%) with 18 wildcards. The 14 files still empty are 10 S-group polymer fixtures, which are left unresolved deliberately, and 4 drawings whose `R` is either undefined in the file or defined only by abbreviations (`RF = SCF3, SeCF3, …`). For the polymer fixtures the other readers return nothing valid: RDKit caps the open ends with hydrogen (`Sgroups_Copolymer 01` becomes `CCC(C)C` + `CCCC`), which yields a valid-looking InChI for a different molecule, while Indigo and ChemScanner leave `*` ends.
- **Indigo is the fastest and the most robust.** 25 ms per file in the cold run and 5.8 ms warm, about 3× faster than BChemXtract 1.4.1 warm (18 ms), no failures on `.cdx`, and no file over 0.1 s. BChemXtract follows at 36 ms with one failure, a file that is not a valid CDX. RDKit is comparable to both on ordinary files (51 ms) but hangs while reading four `volumeTest/bug` files. ChemScanner averages 64 ms without its two slowest files.
- **ChemScanner is unmaintained and breaks on modern files.** Version 0.1.3 dates from 2019 and is the only release. It rejects every CDXML file carrying the current Revvity DTD (12 of 21), because it checks for the CambridgeSoft DTD URL. 32 of its 225 `.cdx` files fail with Ruby exceptions in its interpreter, and 11 of those are argument mismatches against the current RDKit Ruby binding (`Bond.new`). It was run against `rdkit_chem` 2025.09

with one compatibility patch (see Methodology). An RDKit build from 2019 would need a source build and might fail less often.

- **Indigo's CDXML reader is weaker than its CDX reader.** It rejects multi-centre (`multipoint`) attachments ("orphaned node"), Markush alternative groups and multi-arrow schemes, and it crashes with a segfault on Figure1C. That is 7 failures in 21 files.
- **An open BChemXtract gap: Figure1C.** `volumeTest/bug/Figure1C_complete_curated.cdxml` (about 340 aminoglycoside fragments joined by external connection points) yields 0 substances in BChemXtract 1.4.1 and on current `main` . RDKit reads 339 InChIs from it. Its volume test has the assertions commented out, so this is a known open fixture, not a regression.
- **Current `main` with R-groups resolved reaches the same 98.9%.** It reports 1797 unique substances (1777 real InChIs, 19 wildcards) against 1771 for 1.4.1 with R-groups resolved. Two files gain: `bjnano_5_145/m5445236-i5.cdx` goes from 2 to 86, mostly 80 variants of one C_{99} nanographene ($C_{99}H_{59}$–H_{62}) from the cut-mark handling added since 1.4.1, and `StereoInconsistency/m31656373-1.cdx` goes from 10 to 25, with 21 instead of 6 distinct stereo forms of one Ni porphyrin. Markush enumeration yields fewer members than in 1.4.1 (74 against 86 on `.cdxml`), and `vol18Test/m25422595-i12.cdx` goes from 221 to 194. `main` is slower than 1.4.1: 35 against 23 ms per file warm, and the porphyrin file alone takes 3.3 s.
- **BChemXtract 1.4.1 reproduces exactly.** Every count on the `.cdx` set matches the 2026-09-09 report (1360 / 1336 / 24 / 23 / 1337). This run is faster (36 vs 51 ms per file) on the same machine, so absolute times vary with load. Ratios within this report are like for like.

Conclusion

Against today's open-source ChemDraw readers, BChemXtract 1.4.1 keeps a clear lead on extraction **quality**: 98.2% real InChIs (98.9% with R-groups resolved) against 73–77%, and 18–23 unresolved wildcards against 298–378. The margin is about a fifth of the substances rather than Open Babel's sevenfold gap. RDKit and Indigo are capable readers that stop short of resolving ChemDraw labels. On **speed**, Indigo is about 1.4× faster than BChemXtract in the cold run and about 3× faster warm, and the rest are comparable on ordinary files; RDKit has hard hangs on four files. The case for BChemXtract is InChI-grade substance identity, not throughput. The one place it trails is Figure1C, where it extracts nothing and RDKit extracts 339 structures.

Methodology. Same counting rules as the Open Babel report. **BChemXtract** (1.4.1 and `main` at `f1db7bf`): `CDXReader` / `CDXMLReader.readDocument` → `SubstanceXtractor.xtractUnique(doc, info, resolveRGroups)` , with `false` for the plain column and `true` for "+ R-groups", built from tag `v1.4.1` . Version 1.4.1 has no `BCXTractPerformance` , so the current sample was compiled against the 1.4.1 jar. It runs in a single warm JVM. "Real InChI" means a non-empty `getInchi()` ; "unexpanded alias" means SMILES containing `*` . **RDKit**: `Chem.MolsFromCDXMLFile` with `CDXMLFormat.CDX` or `CDXML` (default sanitisation; fragments that fail sanitisation are skipped), then `MolToSmiles` and `MolToInchiKey` . **Indigo**: `iterateCDXFile` for `.cdx` , with reaction objects expanded to their molecules. CDXML is loaded as one molecule (or reaction) and split into connected components. `canonicalSmiles` and IndigoInchi. **ChemScanner**: `ChemScanner::Cdx` / `Cdxml#read` → `molecules.map(&:get_cano_smiles)` , on `rdkit_chem` 2025.09.3.14. Version 0.1.3 passes a detached `Conformer` to stereo perception, which current RDKit rejects ("no owner"). The runner redirects those two calls to the conformer the molecule owns, which holds the coordinates. ChemScanner's Ruby binding has no InChI, so InChIKeys come from Python RDKit applied to its SMILES; this step is not timed. **All toolkits**: one worker process per file, capped at 120 s and 4 GB, with time measured inside the worker and excluding interpreter start-up. "Unique" means distinct InChIKeys plus structures without an InChI, deduplicated on SMILES. Open Babel was rerun the same day with `testset/run-openbabel-performance.sh` ; its counts match the September report. Reproduce with `testset/toolkits/run-toolkit-performance.py {rdkit|indigo|chemscanner} DIR --ext {cdx|cdxml}` . The machine is an Intel Core i5-12500H with 16 threads running Linux Mint 22.3, with Temurin 25.0.4.1+1 and Python 3.12.